

Supplementary Material

When AI Exposure Rankings Disagree: Robust Top-25 Selection Across 661 U.S. Occupations

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S1. Input and scenario register

The decision layer uses three frozen normalized exposure channels: AI Occupational Exposure (AIOE) [1], Microsoft user-goal, and Microsoft AI-action/nonphysical [2]. Employment scale is May 2025 OEWS employment share [3], and the five occupation-level skill descriptors are O*NET 30.3 Importance values [4]. The construction follows transparent composite-indicator practice [5].

Let $a_i^{(r)} \in [0, 1]$ denote frozen normalized exposure for channel r and $z_{im} \in [0, 1]$ the frozen normalized O*NET value for skill m . The five skills are Critical Thinking, Active Learning, Learning Strategies, Active Listening, and Speaking. For construction w with retained set M_w ,

$$h_i^{(w)} = \frac{1}{|M_w|} \sum_{m \in M_w} z_{im}, \quad q_i^{(r,w)} = a_i^{(r)} (1 - h_i^{(w)}), \quad p_i^t = s_i q_i^{(r,w)}.$$

The factor $1 - h_i^{(w)}$ is a declared screening transformation: a larger retained-skill aggregate lowers the modifier within this construction, but this does not establish a causal protective effect of skills. The equal-five construction uses all five skills; each of the five leave-one-out constructions removes exactly one skill. Thus the 18 primitive priority vectors are the Cartesian product of three exposure channels and six constructions. The upstream normalized values are treated as frozen inputs to this decision-layer paper and are not re-estimated here. The priority box is $\ell_i = \min_t p_i^t$, $u_i = \max_t p_i^t$; the point core is $c_i = p_i^{\text{AIOE, equal-five}}$, hence $\ell_i \leq c_i \leq u_i$.

Table 1: Primitive scenario construction.

Axis	Admitted levels
Exposure channel	AIOE; Microsoft user-goal; Microsoft AI-action/nonphysical
Skill construction	equal-five; leave Critical Thinking out; leave Active Learning out; leave Learning Strategies out; leave Active Listening out; leave Speaking out

S2. Membership counts and the 14 stable occupations

Table 2: Membership counts by selection size.

k	Scenario intersection	Scenario union	Box guaranteed	Box possible
10	7	16	2	43
25	14	37	4	87
50	32	73	14	159
100	68	135	27	244

At $k = 25$, the 14 occupations present in every coherent Top-25 are:

1. General and Operations Managers (11-1021);
2. Accountants and Auditors (13-2011);
3. Registered Nurses (29-1141);
4. Waiters and Waitresses (35-3031);
5. Janitors and Cleaners, Except Maids and Housekeeping Cleaners (37-2011);
6. Cashiers (41-2011);
7. Retail Salespersons (41-2031);
8. Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products (41-4012);

9. Bookkeeping, Accounting, and Auditing Clerks (43-3031);
10. Receptionists and Information Clerks (43-4171);
11. Secretaries and Administrative Assistants, Except Legal, Medical, and Executive (43-6014);
12. Office Clerks, General (43-9061);
13. Heavy and Tractor-Trailer Truck Drivers (53-3032);
14. Stockers and Order Fillers (53-7065).

These are stable screening priorities within the declared 18-specification family; the result is not a displacement or program recommendation.

S3. Exact rank proof and tie convention

Best rank is existential:

$$r_i^{\min} = 1 + \sum_{j \neq i} \mathbf{1}\{\ell_j > u_i\}.$$

Worst rank is pessimistic:

$$r_i^{\max} = 1 + \sum_{j \neq i} \mathbf{1}\{u_j \geq \ell_i\}.$$

Here $\mathbf{1}\{\cdot\}$ denotes the indicator function. Possible membership is $r_i^{\min} \leq k$ and guaranteed membership is $r_i^{\max} \leq k$. The strict/non-strict distinction controls boundary inclusion in the contraction analysis.

S4. Safe-screening proof detail

Let $G = \{i : r_i^{\max} \leq k\}$ and $Z = \{i : r_i^{\min} > k\}$. Order the items in G by nonincreasing lower bound and insert them in that order. If the next $a \in G$ is absent from a size- k optimum, at most $k - 1$ competitors satisfy $u_j \geq \ell_a$, so the list contains an h with $u_h < \ell_a$. Any guaranteed item a' inserted earlier has $\ell_{a'} \geq \ell_a$ and hence $u_{a'} \geq \ell_{a'} \geq \ell_a$, so it cannot be chosen as h . Replacing h by a therefore preserves all earlier guaranteed insertions and weakly increases every scenario-wise selected mass.

After all G items have been inserted, suppose a selected $i \in Z$ remains. At least k competitors satisfy $\ell_j > u_i$, so at least one such j is omitted. Replacing i by that j strictly increases every scenario-wise selected mass. Because the family of size- k lists is finite, strict exchanges cannot cycle. If a Z item remained, another strict exchange would exist; therefore the process terminates at an optimum containing all G and no Z . This establishes the existence claim for every objective coordinatewise nonincreasing in scenario-wise selected mass.

S5. Fixed-list verification, consensus benchmark, and criterion sensitivity

Table 3: Fixed-list comparison at $k = 25$ over 18 primitive scenarios.

Rule	Mean capture	Minimum capture	Max. rel. regret	Box regret
Scale only	93.02%	88.35%	11.65%	0.02380
Baseline composite	96.14%	93.70%	6.30%	0.01810
Consensus-Frequency	97.00%	95.89%	4.11%	0.01952
Relative minimax	96.99%	96.52%	3.48%	0.01945

For the Consensus-Frequency benchmark, let $T_t = \text{Top}_{25}(p^t)$ denote the 25 occupations with the largest priorities under scenario t . Then $f_i = \sum_{t=1}^{18} \mathbf{1}\{i \in T_t\}$ counts appearances across the 18 scenario-specific Top-25 lists. The 25 largest counts are selected; the cutoff is strict (9 appearances for the 25th occupation versus 8 for the 26th), so no frequency tie occurs at the membership cutoff. Ascending SOC is retained only as a fixed deterministic tie-break for equal counts away from the cutoff. The list overlaps relative minimax in 24/25 occupations: consensus includes Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031), whereas minimax includes Amusement and Recreation Attendants (39-3091). Exact scenario-by-scenario capture/regret are retained in the reproducibility addendum. Counts are design occurrences, not probabilities.

Consensus-Frequency Top-25 SOC membership. 11-1021, 13-2011, 29-1141, 35-3031, 37-2011, 41-2011, 41-2031, 41-4012, 43-3031, 43-4171, 43-6014, 43-9061, 53-3032, 53-7065, 13-1111, 43-1011, 13-1071, 15-1232, 33-9032, 35-2014, 35-9031, 41-1011, 43-5071, 43-6013, and 49-9071. The complete labeled list is supplied in `consensus_frequency_selection.csv`. The relative-minimax list contains the same 24 shared occupations, replacing Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031) with Amusement and Recreation Attendants (39-3091).

The relative-minimax objective is $\max_t R_t(x)$ with $R_t(x) = 1 - \sum_i p_i^t x_i / V_t^*$ and $V_t^* > 0$ for every admitted scenario. The absolute-minimax sensitivity instead defines

$$R_t^{\text{abs}}(x) = V_t^* - \sum_i p_i^t x_i, \quad x: \min_{x_i=k, x_i \in \{0,1\}} \max_t R_t^{\text{abs}}(x).$$

Its list overlaps the relative-minimax list in 23/25 occupations, with minimum relative capture 94.79% and maximum relative regret 5.21%.

S6. Nested-support sensitivity and exact boundary semantics

For retained width $g \in [0, 1]$,

$$[L_i(g), U_i(g)] = [c_i - g(c_i - \ell_i), c_i + g(u_i - c_i)].$$

We denote each reported structural transition boundary by g^* . These values are transition boundaries. Under the stated tie

Table 4: Structural Top-25 transition boundaries.

Event	g^*	$1 - g^*$	Identity check
Guaranteed $\rightarrow$ 14	0.5743200867	0.4256799133	coherent-intersection set after crossing
Possible $\rightarrow$ 37	0.4346510223	0.5653489777	cardinality only; 32 shared
Full 25/25	0.0787564209	0.9212435791	point list

convention, the post-contraction state applies for $g < g^*$ when contraction proceeds downward from $g = 1$; equality is evaluated by the original strict $>$ and non-strict $\geq$ rules rather than by rounded prose labels.

At the 37-member boundary, the five possible-set-only occupations are Computer and Information Systems Managers (11-3021), Medical Assistants (31-9092), Billing and Posting Clerks (43-3021), First-Line Supervisors of Construction Trades and Extraction Workers (47-1011), and Inspectors, Testers, Sorters, Samplers, and Weighers (51-9061). The five coherent-union-only occupations are Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop (35-9031), Amusement and Recreation Attendants (39-3091), Counter and Rental Clerks (41-2021), Production, Planning, and Expediting Clerks (43-5061), and First-Line Supervisors of Production and Operating Workers (51-1011).

For a strictly decreasing homeomorphism $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 1$, $\phi(1) = 0$, set $g = \phi(\alpha)$. Linear $\phi(\alpha) = 1 - \alpha$ and quadratic $\phi(\alpha) = (1 - \alpha)^2$ relabel the same support family and cannot constitute independent robustness tests.

S7. Claim-admissibility matrix

Table 5: Permitted and unsupported interpretations.

Output	Permitted	Unsupported
Box guaranteed/possible	Exact membership under independent priority intervals	Probability of selection
Coherent intersection/union	Membership over 18 admitted designs	Universal robustness
Relative-minimax list	Best worst relative capture over declared scenarios	Optimal workforce policy
Consensus-frequency list	Occurrence-based aggregation over 18 admitted Top-25 lists	Probability of selection or worst-case optimum
Box regret	Exact absolute regret under independent priority intervals	Dominance under coherent designs
Boundary g^*	Width at which rank-comparison status can change	Confidence or calibrated possibility

References

- [1] E. W. Felten, M. Raj, and R. Seamans, “Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses,” *Strategic Management Journal*, vol. 42, no. 12, pp. 2195–2217, 2021, doi: 10.1002/smj.3286.
- [2] K. Tomlinson, S. Jaffe, W. Wang, S. Counts, and S. Suri, “Working with AI: Measuring the Applicability of Generative AI to Occupations,” arXiv:2507.07935, 2025, doi: 10.48550/arXiv.2507.07935.
- [3] U.S. Bureau of Labor Statistics, *Occupational Employment and Wage Statistics, May 2025 National Estimates*, 2026.
- [4] U.S. Department of Labor, Employment and Training Administration, *O*NET 30.3 Database*, May 2026 release.
- [5] OECD and Joint Research Centre, *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, 2008.